

Lesson 1: Def, Thm, Proof

Connectives : "not", and, or, if...then
if and only if.

Precise meaning: goal

Everyday usage: ambiguous

Lesson 2 Logical connectives

"not" operates on a statement to
produce the negation of the statement

P : "x is an even integer"
↑
statement
(x=2)
↑
variable

negation of P : $\sim P$ ← input statement
not P

Truth table for "not"
"x is not an even integer."

P	$\sim P$
T	F
F	T

• $a > b$

negation : $a \not> b$

$a \leq b$

• it does not rain ←
negation : it rains

← slash = "not"

$$\sim(\sim P) = P$$

"and" : binary operation $\rightarrow$ it takes in
~~"but"~~ 2 statements and produces
 a conjunction

p and q

$p \wedge q$

$\nwarrow \nearrow$
 input statements

p : x is an even integer

q : x is a positive integer

$p \wedge q$: x is a positive even integer
 x is a positive and even integer

$$p \wedge q = q \wedge p$$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$= 2 \times 2$
 4 possible
 combinations
 of truth
 values for
 p and q

"or" : binary operation

p or q : disjunction
 $p \vee q$

Ex p, q same as before

$p \text{ or } q$: x is an even integer or
 $p \vee q$ a positive integer

p	q	$p \vee q$	"inclusive"
T	T	T	
T	F	T	
F	T	T	
F	F	F	

$$p \text{ or } q = p \vee q \text{ (or both)}$$

Everyday : ambiguous

- Can be exclusive :

We will have Thai or pizza
for lunch (not both)

- Can be inclusive :

passport or driver's license
as your ID.

"if...then"

Ex: If it does not rain, then I go jogging
 $\underbrace{\hspace{10em}}_p \quad \underbrace{\hspace{10em}}_q$

if p then q : conditional(statement)

$$p \rightarrow q$$

p implies q

premise/
antecedent/
hypothesis

conclusion/
consequent

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

$p \rightarrow q \neq \sim p \rightarrow \sim q$ inverse

$p \rightarrow q \neq q \rightarrow p$ converse

p	q	$p \rightarrow q$	$q \rightarrow p$	$\sim p$	$\sim q$	$\sim p \rightarrow \sim q$
T	T	T	T	F	F	T
T	F	F	F	F	T	F
F	T	T	T	T	F	T
F	F	T	F	T	T	T

$\downarrow T \rightarrow F = F$

$\downarrow F \rightarrow T = T$

if and only if

$$P \leftrightarrow Q = (P \rightarrow Q) \wedge (Q \rightarrow P)$$

iff
biconditional

$$(P \rightarrow Q) \wedge (\sim P \rightarrow \sim Q)$$

($\wedge$)

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$P \leftrightarrow Q$
T	T	T	T	T
T	F	F	F	F
F	T	T	F	F
F	F	T	T	T

Def

x is even (if) $x = 2n$
for some integer n

if and only if

x is even $\leftrightarrow x = 2n$
... integer